

Wave 5.2R Stage 15 Official Forward Verification And Deployment Preparation

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Executive Decision

Stage 15 is complete on the official polished-dataset setpoint-forward surface. The matrix evaluated 97 held-out forward curves with the exact same split and runtime-valid input contract for H04, PF-A, the accepted harmonic MLP, and the accepted periodic GRU.

H04 remains exploratory and receives no family or program registry promotion. It improves PF-A raw MAE by 4.587% and is the best candidate for mean-centered shape, derivative fidelity, and mean harmonic phase. It does not displace the GRU: its raw MAE is 6.814% worse and its absolute offset error is 28.116% worse. The GRU also retains the best P95 error and peak-to-peak behavior.

The Python/ONNX and independent float32 PLC-reference parity gates pass. A TwinCAT compile and runtime replay were not performed, so this report makes no PLC runtime claim. The accepted periodic GRU remains the forward incumbent.

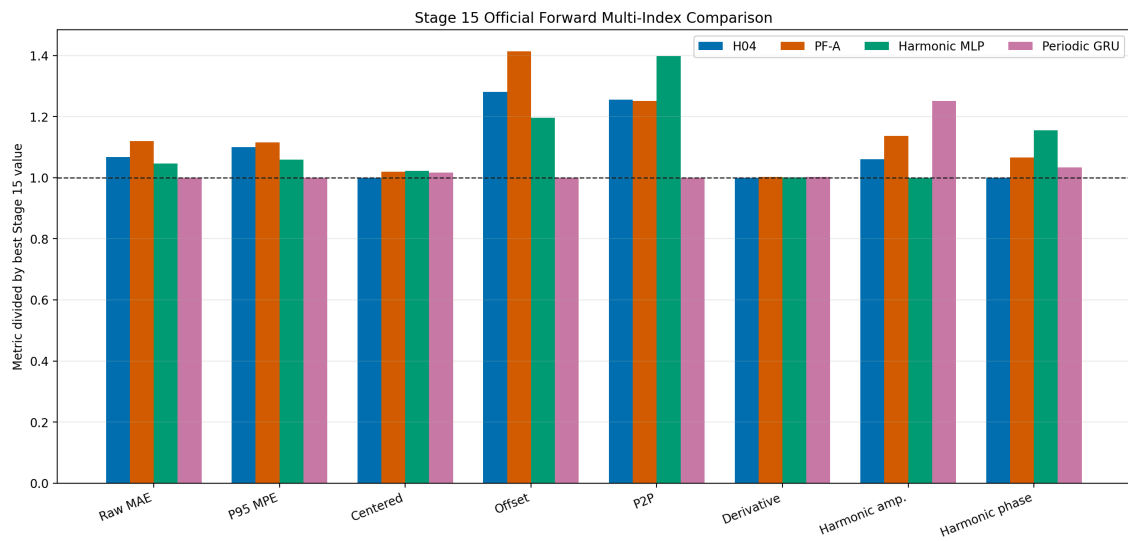
Scope And Evidence Contract

- dataset: data/polished_dataset ;
- surface: setpoint Fw only;
- held-out curves: 97 ;
- angular samples per full curve: 2048 ;
- candidates: H04, PF-A, accepted harmonic MLP, accepted periodic GRU;
- selection policy: multi-index curve-first, never scalar MAE alone;
- curve payload diagnostics: CVP 1.2 with full curves and downsampled visual payloads;
- deployment evidence: immutable Python checkpoint replay, ONNX parity, and static Structured Text reference parity.

Official Multi-Index Results

Metric	H04	PF-A	MLP	GRU	Best
Raw MAE [deg]	0.0017286	0.0018118	0.0016935	0.0016184	Periodic GRU
RMSE [deg]	0.0020372	0.0021259	0.0020077	0.0019306	Periodic GRU
MPE [%]	3.5603777	3.7486325	3.4385998	3.2784267	Periodic GRU
P95 MPE [%]	7.9470835	8.0612904	7.6516177	7.2247095	Periodic GRU
Centered shape MAE [deg]	0.0013590	0.0013850	0.0013900	0.0013820	H04

Metric	H04	PF-A	MLP	GRU	Best
Absolute offset error [deg]	0.0008840	0.0009750	0.0008250	0.0006900	Periodic GRU
Peak-to-peak error [%]	10.6001280	10.5672770	11.8017840	8.4436780	Periodic GRU
Derivative RMSE [deg/deg]	0.0149380	0.0149690	0.0149630	0.0149680	H04
Harmonic amplitude error [%]	16.3185540	17.4753110	15.3800510	19.2368600	Harmonic MLP
Harmonic phase error [deg]	11.7719440	12.5529950	13.5916490	12.1657890	H04
Closure mismatch [deg]	0.0008690	0.0008690	0.0008810	0.0008920	H04

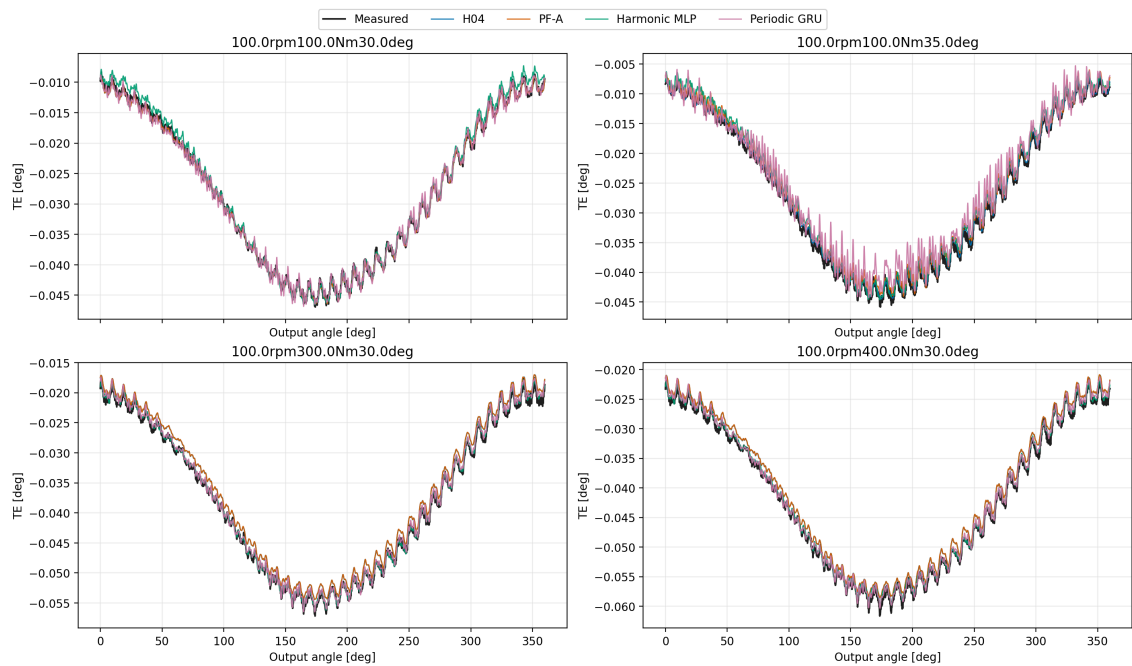


Stage 15 official forward multi-index comparison

The official matrix and CVP 1.2 diagnostics separate the winners:

- best raw MAE, RMSE, MPE, P95 MPE, offset, and peak-to-peak behavior: [periodic GRU](#);
- best centered shape, derivative fidelity, and mean harmonic phase: [H04](#);
- best aggregate CVP 1.2 diagnostic score and harmonic amplitude error: [harmonic MLP](#);
- PF-A is improved by H04 but is not the official winner on any decisive acceptance axis. H04's shape improvement over the GRU is [1.664%](#) . This confirms that the bounded analytical residual learned useful periodic structure. The simultaneous raw and offset regressions show that this advantage is not a balanced replacement for the incumbent.

Representative Curve Evidence



Stage 15 representative forward overlays

The overlays use four deterministic CVP 1.2 payload conditions. They are visual evidence only; the decision above uses all 97 full-resolution curves. All models reproduce the dominant periodic structure. Their separation is small enough that scalar or hand-selected visual inspection alone would be misleading, which is why the official decision retains the multi-index table.

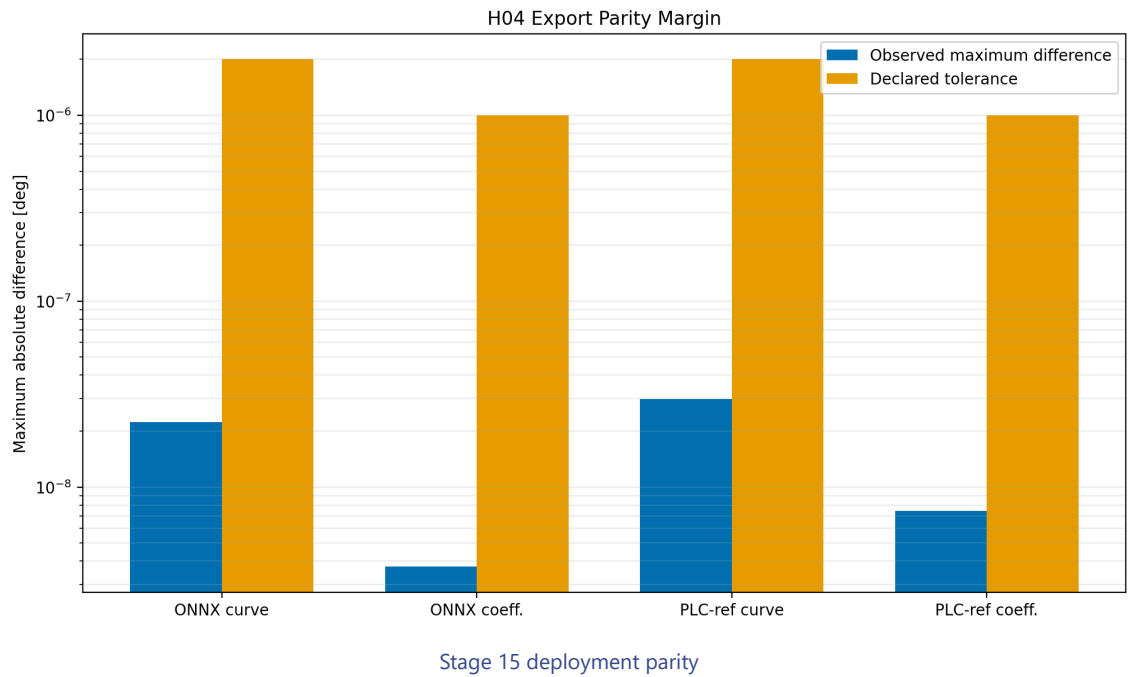
Robustness Interpretation

The temperature slices preserve the same overall conclusion:

Temperature	H04 MAE [deg]	PF-A MAE [deg]	MLP MAE [deg]	GRU MAE [deg]
25 C	0.0013770	0.0014953	0.0012815	0.0013628
30 C	0.0018189	0.0018381	0.0017768	0.0016156
35 C	0.0018864	0.0020149	0.0019034	0.0018093

H04 improves PF-A at every available temperature. It does not establish a uniform advantage over the accepted neural references: the MLP leads at 25 C, while the GRU leads at 30 C and 35 C. The result supports retaining H04 as a compact grey-box formulation, not promoting it as the balanced forward incumbent.

Deployment Preparation And Parity



Check	Observed maximum difference	Tolerance	Result
ONNX reconstructed curve	2.235174179e-08 deg	2.0e-06 deg	pass
ONNX final coefficients	3.725290298e-09 deg	1.0e-06 deg	pass
PLC-reference reconstructed curve	2.980232239e-08 deg	2.0e-06 deg	pass
PLC-reference final coefficients	7.450580597e-09 deg	1.0e-06 deg	pass

The export package contains the ONNX graph, immutable parity payload, Structured Text function block, Structured Text parameter GVL, and the parameter archive. This proves reproducible static numerical translation. TwinCAT compilation, execution-time measurement, task integration, invalid input behavior, saturation behavior, and online `DataValid` replay remain future deployment work.

Acceptance Gate

Gate	Result
Official common forward matrix completed	pass
H04 improves PF-A raw MAE	pass
H04 improves incumbent raw MAE	fail
H04 improves incumbent centered shape	pass
H04 improves incumbent offset	fail
Python/ONNX parity	pass
Static PLC-reference parity	pass
TwinCAT compile and runtime parity	pending / not claimed

The Stage 15 exit gate therefore closes without promotion. This is a valid negative acceptance result: the challenger demonstrated real shape value and excellent export parity, but it did not provide a balanced predictive gain and does not yet have runtime PLC evidence.

Program Decision

Wave 5.2R has completed all stages 0 through 15. The program conclusions are:

1. observable analytical and harmonic priors are useful when treated as bounded structure, diagnostics, or auxiliary objectives;
2. useful physics guidance does not guarantee a better balanced predictor;
3. H04 is the strongest compact grey-box output of the wave and should be preserved for future deployment research;
4. the periodic GRU remains the accepted forward incumbent;
5. no family or program registry changes are authorized by this result;
6. physics-integrated Wave 6 remains a separate future decision and must not inherit an unearned H04 acceptance claim.

Reproducibility Artifacts

- official matrix:
`output/validation_checks/track2_reference_comparison/2026-07-30-01-03-11__wave52r_stage15_official_forward_verification_wave52r_stage15_official_forward_verification/ ;`
- CVP 1.2 diagnostics:
`output/validation_checks/wave52r_stage15_curve_payload_diagnostics/2026-07-30-01-11-50__track2c_curve_payload_diagnostics/ ;`
- deployment parity:
`output/validation_checks/wave52r_stage15_deployment_parity/ ;`
- machine-readable decision:
`output/analysis/wave_5_2r/stage15_official_forward_verification/closeout/stage15_official_forward_verification_decision.yaml .`